

Daily Gas Usage and Prediction



Company Background

Scot Forge, headquartered in Spring Grove, Illinois, is an employee-owned manufacturer specializing in custom open-die and rolled-ring forgings. With more than a century of experience, the company supplies critical components to aerospace, defense, mining, semiconductor manufacturing, and power generation. **Scot Forge** is recognized for producing high-quality, complex parts and strong technical capability. For more information, visit www.scotforge.com.

Problem Background

Many of **Scot Forge's** operations require significant natural gas use, especially in furnaces used for forging and heat treatment. Recent Illinois legislation changed how natural gas tariffs are calculated, adding limits that can trigger penalties. These penalties depend on daily injections or withdrawals from the company's natural-gas storage bank and the total amount stored at the end of each month.

Both the Spring Grove and Franklin Park facilities are affected. Across the two plants, more than 60 furnaces may be active, but furnace-level gas usage is not currently measured. Instead, **Scot Forge** receives facility-level meter readings from a third-party provider. These readings are used to estimate usage, relying heavily on intuition and informal historical comparisons. This dataset will serve as the basis for your analysis.

Problem Statement

Your task is to prepare and analyze the provided natural-gas usage data and develop a time-series model that predicts next-day usage. This includes cleaning the dataset, exploring key patterns, and using historical behavior to inform the forecast.

Objectives

1. **Exploratory Data Analysis (EDA):** Review the dataset, identify key patterns and anomalies, and create visualizations that clearly communicate findings.
 2. **Research Questions:** Develop focused research questions and hypotheses to guide the analysis.
 3. **Hypothesis Testing:** Apply statistical tests to evaluate each hypothesis and determine whether to reject or fail to reject the null.
 4. **Draft Report (MVP):** Produce a mid-term summary of initial insights supported by clear visuals.
 5. **Data Preparation:** Clean and structure the data by correcting issues, handling missing values, aligning timestamps, and engineering useful features.
 6. **Baseline Model:** Build and document a simple forecasting model that reflects **Scot Forge's** current estimation method.
 7. **Forecasting Model:** Develop a time-series model to predict next-day gas usage and explain the factors that influence predictions.
 8. **Model Evaluation:** Assess both models using appropriate time-series metrics and visuals that highlight performance and errors.
 9. **Final Report:** Deliver a formal report summarizing the full analysis, modeling, evaluation, and recommendations.
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The **Scot Forge** team will meet with you throughout the project to review progress. Treat them as business experts rather than coding experts. Clearly explain your findings, the reasoning behind your decisions, and the implications for the business. You will also be expected to manage meetings, deadlines, deliverables, and communication in a professional manner.